Human review packet: GS62CollegeAdmissions

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GS62CollegeAdmissions
Paper id: `GS62CollegeAdmissions`
Source version: American Mathematical Monthly 69(1), January 1962, printed pages 9-15
Source record: `audit/paper_statement_map.json`
Rows: 5
Generated: 2026-08-18
Source URL: official source

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GS62CollegeAdmissions --serve
```

Open `http://127.0.0.1:8765/` in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local review trace.

Contents

  - GS62CollegeAdmissions.PaperInterface.unstableMarriage
- Main-text source claims (5)
  - unstableCollegeAssignment_iff_source_definitionSpec
  - literalApplicantOptimalCollegeAssignment_iff_source_definitionSpec
  - unstableMarriage_iff_source_definitionSpec
  - theorem1_stable_marriage_existsSpec
  - theorem2_applicant_optimalitySpec

Material library prerequisites

`EconCSLib.FiniteChoice.linearTopQChoice`

Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   THEOREM 2. Everyapplicant is at least as well offundertheassignmentgiven
by thedeferredacceptanceprocedureas he would be underany otherstableassign-
ment.

[Next verbatim source excerpt]

   DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
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   5. Optimality.We now show that the "deferredacceptance" procedurejust
described yields not only a stable but an optimal assignment of applicants.
That is,
```

Lean-expanded library semantic target

```
{α : Type u_1} →
  [inst : DecidableEq α] →
    [inst_1 : LinearOrder α] →
      (q : ℕ) →
        (a : Finset α) →
          if h : q ≤ a.card then Finset.image (fun i => ↑((a.orderIsoOfFin ⋯) ⟨↑i, ⋯⟩)) Finset.univ else a
```

Exact Lean library declaration

```
noncomputable def linearTopQChoice (q : ℕ) : ChoiceRule α :=
  fun X =>
    if h : q ≤ X.card then
      Finset.univ.image
        (fun i : Fin q =>
          ((X.orderIsoOfFin (by rfl)
            (i.1, lt_of_lt_of_le i.2 h) : X) : α))
    else
      X
```

Recorded source-to-library screening **Verdict:** `matches`

Reason: The source says each college retains the q applicants it ranks highest, or all applicants if fewer apply. The Lean library function is exactly that top-q selection from the candidate pool under the college order.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`Assignment M W`

Source locator: source.txt:133-139 (printed page 11, Section 3)

Verbatim paper-source connection

```
A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Lean declaration type (metadata)

```
Type u_1 → Type u_2 → Type (max u_1 u_2)
```

Exact Lean library declaration

```
/-- A matching between Men and Women. -/
structure Assignment (M W : Type*) where
  m_match : M → Option W
  w_match : W → Option M
  consistent_m : ∀ m w, m_match m = some w ↔ w_match w = some m
```

Recorded source-to-library screening **Verdict:** `matches`

Reason: The source's set of marriages pairs each participant with an actual mate. The Lean structure records both partner maps and their mutual consistency; the PaperInterface predicate separately states the source's complete-marriage domain.

Reviewer check: ☐ Matches source input

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Reviewer annotation

`ManyToOne.valApplicant`

Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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```

Lean-expanded library semantic target

```
{Applicants : Type u_1} →
  {Colleges : Type u_2} →
    (val_applicant : Applicants → Colleges → ℝ) →
      (a : Applicants) →
        (c : Option Colleges) →
          match c with
          | none => 0
          | some c' => val_applicant a c'
```

Exact Lean library declaration

```
def valApplicant {Applicants Colleges : Type*}
    (val_applicant : Applicants → Colleges → ℝ)
    (a : Applicants) (c : Option Colleges) : ℝ :=
  match c with
  | none => 0
  | some c' => val_applicant a c'
```

Recorded source-to-library screening **Verdict:** `matches`

Reason: Sections 4–5 make a terminal applicant either waitlisted or rejected by every willing and permitted college. The procedure's applicant comparison uses an explicit zero outside option for that unmatched terminal case; when matched, the definition returns exactly the applicant's value for the assigned college.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`ManyToOneAssignment Applicants Colleges`

Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim paper-source connection

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                    A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Lean declaration type (metadata)

```
Type u_1 → Type u_2 → Type (max u_1 u_2)
```

Exact Lean library declaration

```
/-- A many-to-one assignment between applicants and colleges. -/
structure ManyToOneAssignment (Applicants Colleges : Type*) where
  app_match : Applicants → Option Colleges
  college_roster : Colleges → Finset Applicants
  consistent : ∀ a c, app_match a = some c ↔ a ∈ college_roster c
```

Recorded source-to-library screening Verdict: `matches`

Reason: The source describes applicants assigned to colleges and later permits an applicant to be rejected by every college willing to consider them. The Lean structure records the applicant-to-college relation, college rosters, and their consistency, including that permitted unmatched outcome.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`valM`

Source locator: source.txt:133-139 (printed page 11, Section 3)

Verbatim paper-source connection

```
A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    (val : M → W → ℝ) →
      (m : M) →
        (w : Option W) →
          match w with
          | none => 0
          | some w' => val m w'
```

Exact Lean library declaration

```
/-- The value of a match for a man. 0 if unmatched. -/
def valM {M W : Type*} (val : M → W → ℝ) (m : M) (w : Option W) : ℝ :=
  match w with
  | none => 0
  | some w' => val m w'
```

Recorded source-to-library screening **Verdict:** `matches`

Reason: On the explicitly complete marriage domain of the reviewed source claim, the only relevant branch returns the source man's value for his actual mate; the none-to-zero branch does not represent a source actual mate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`valW`

Source locator: source.txt:133-139 (printed page 11, Section 3)

Verbatim paper-source connection

```
A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
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munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    (val : W → M → ℝ) →
      (w : W) →
        (m : Option M) →
          match m with
          | none => 0
          | some m' => val w m'
```

Exact Lean library declaration

```
/-- The value of a match for a woman. 0 if unmatched. -/
def valW {M W : Type*} (val : W → M → ℝ) (w : W) (m : Option M) : ℝ :=
  match m with
  | none => 0
  | some m' => val w m'
```

Recorded source-to-library screening **Verdict:** `matches`

Reason: On the explicitly complete marriage domain of the reviewed source claim, the only relevant branch returns the source woman's value for her actual mate; the none-to-zero branch does not represent a source actual mate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-specific semantic prerequisites

These paper-local state, policy, or transition declarations remain named in the Lean-expanded claim because they are independent semantic objects. Each is shown with its own verbatim source connection and reviewer annotation before the claim that uses it. They are prerequisites, not extra paper-claim rows.

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState`
Source locator: source.txt:232-249

Verbatim paper-source connection

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```

Lean-elaborated paper signature

```
(Applicants : Type u_3) → Type u_4 → [DecidableEq Applicants] → Type (max u_3 u_4)
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: Sections 4–5 describe applicants applying in rounds, colleges retaining a quota-sized waiting list, and the terminal lists determining assignments. The Lean state records exactly the per-college waiting lists needed for that source procedure.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.collegeOrder`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.collegeOrder`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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That is,
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Lean-expanded paper semantic target

```
{Applicants : Type u_1} →
  {Colleges : Type u_2} →
    (val_college : Colleges → Applicants → ℝ) →
      (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
        (c : Colleges) → LinearOrder.lift' (fun a => OrderDual.toDual (val_college c a)) ⋯
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: Each college retains the applicants it ranks highest. The Lean order is the strict college ranking used to make that top-q choice.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.collegeTopQ`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.collegeTopQ`
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    [inst : DecidableEq Applicants] →
      (quota : Colleges → ℕ) →
        (val_college : Colleges → Applicants → ℝ) →
          (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
            (c : Colleges) → (pool : Finset Applicants) → EconCSLib.FiniteChoice.linearTopQChoice (quota c) pool
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: The source says a college places its q highest-ranked applicants, or all applicants if fewer, on its waiting list. The Lean function is that literal top-q selection.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.waitingList`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.waitingList`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

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[Next verbatim source excerpt]

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That is,
```

Lean-expanded paper semantic target

```
{Applicants : Type u_1} →
  {Colleges : Type u_2} →
    [inst : DecidableEq Applicants] →
      (quota : Colleges → ℕ) →
        (val_college : Colleges → Applicants → ℝ) →
          (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
            (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
              (c : Colleges) →
                GS62CollegeAdmissions.ExactCollegeBatchedProcedure.collegeTopQ quota val_college hcollegeStrict c
                  (s.applications c)
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: The source repeatedly forms a college waiting list by retaining its highest-ranked applicants among the old list and new applications. The Lean definition is that list at a procedure state.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.assignedCollege`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.assignedCollege`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

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   DEFINITION. An assignment  ofapplicantstocolleges
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thereare twoapplicantsa and ,Bwhoare assignedtocolleges
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    [inst : Fintype Colleges] →
      [inst_1 : DecidableEq Applicants] →
        (quota : Colleges → ℕ) →
          (val_college : Colleges → Applicants → ℝ) →
            (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
              (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
                (a : Applicants) →
                  if h :
                      ∃ c,
                        a ∈
                          GS62CollegeAdmissions.ExactCollegeBatchedProcedure.waitingList quota val_college
                            hcollegeStrict s c then
                    some (Classical.choose h)
                  else none
```

Recorded source-to-declaration screening **Verdict:** `matches`
Reason: At termination the source admits each applicant on a college waiting list and leaves every other applicant unassigned. The Lean function reads that assignment from the lists.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceEligible`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceEligible`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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Lean-expanded paper semantic target

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  (val_college : Colleges → Applicants → ℝ) (a : Applicants) (c : Colleges), 0 < val_applicant a c ∧ 0 < val_college c a
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: The source excludes colleges unwilling to accept a student and applicants not willing or permitted to apply. The Lean predicate records precisely the mutual eligibility condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.untriedEligibleColleges`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.untriedEligibleColleges`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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```
{Applicants : Type u_1} →
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    [inst : Fintype Colleges] →
      [inst_1 : DecidableEq Applicants] →
        (val_applicant : Applicants → Colleges → ℝ) →
          (val_college : Colleges → Applicants → ℝ) →
            (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
              (a : Applicants) →
                {c |
                  GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceEligible val_applicant val_college a c ∧
                    a ∉ s.applications c}
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: The procedure continues only while an applicant has a college to which the applicant is willing and permitted to apply and has not already tried. The Lean set is that remaining choice set.
Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceActive

Paper-local declaration:
GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceActive
Source locator: source.txt:232-249

Verbatim paper-source connection

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Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} [inst : Fintype Colleges] [inst_1 : DecidableEq Applicants]
  (quota : Colleges → ℕ) (val_applicant : Applicants → Colleges → ℝ) (val_college : Colleges → Applicants → ℝ)
  (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a')
  (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) (a : Applicants),
  GS62CollegeAdmissions.ExactCollegeBatchedProcedure.assignedCollege quota val_college hcollegeStrict s a = none ∧
    (GS62CollegeAdmissions.ExactCollegeBatchedProcedure.untriedEligibleColleges val_applicant val_college s a).Nonempty
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The source stops when no applicant remains able to apply to a college that would consider them. The Lean predicate identifies that remaining active/applicable condition, so its negation is the source stopping condition used in the theorem.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GS62CollegeAdmissions.ExactCollegeBatchedProcedure.nextCollege

Paper-local declaration:
GS62CollegeAdmissions.ExactCollegeBatchedProcedure.nextCollege
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

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That is,
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Lean-expanded paper semantic target

```
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    [inst : Fintype Applicants] →
      [inst_1 : Fintype Colleges] →
        [inst_2 : DecidableEq Applicants] →
          [inst_3 : DecidableEq Colleges] →
            (quota : Colleges → ℕ) →
              (val_applicant : Applicants → Colleges → ℝ) →
                (val_college : Colleges → Applicants → ℝ) →
                  (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
                    (s :
                        GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
                      (a : Applicants) →
                        if hactive :
                            GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceActive quota val_applicant
                              val_college hcollegeStrict s a then
                          some (Classical.choose ⋯)
                        else none
```

Recorded source-to-declaration screening **Verdict:** `matches`
Reason: After rejection, the source directs an applicant to the next eligible choice. The Lean function selects that next untried eligible college in the applicant's preference order.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.ExactCollegeBatchedProcedure.newApplications

Paper-local declaration:
GS62CollegeAdmissions.ExactCollegeBatchedProcedure.newApplications
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

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            (quota : Colleges → ℕ) →
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                (val_college : Colleges → Applicants → ℝ) →
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                    (s :
                        GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
                      (c : Colleges) →
                        {a |
                          GS62CollegeAdmissions.ExactCollegeBatchedProcedure.nextCollege quota val_applicant val_college
                              hcollegeStrict s a =
                            some c}
```

Recorded source-to-declaration screening **Verdict:** `matches`
Reason: The source sends each rejected applicant to the next college on that applicant's list that is willing and permitted to consider the applicant. The Lean function collects exactly those round applications.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceStep

Paper-local declaration:
GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceStep
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

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will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   5. Optimality.We now show that the "deferredacceptance" procedurejust
described yields not only a stable but an optimal assignment of applicants.
That is,
```

Lean-expanded paper semantic target

```
{Applicants : Type u_1} →
  {Colleges : Type u_2} →
    [inst : Fintype Applicants] →
      [inst_1 : Fintype Colleges] →
        [inst_2 : DecidableEq Applicants] →
          [inst_3 : DecidableEq Colleges] →
            (quota : Colleges → ℕ) →
              (val_applicant : Applicants → Colleges → ℝ) →
                (val_college : Colleges → Applicants → ℝ) →
                  (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
                    (s :
                        GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
                      {
                        applications := fun c =>
                          s.applications c ∪
                            GS62CollegeAdmissions.ExactCollegeBatchedProcedure.newApplications quota val_applicant
                              val_college hcollegeStrict s c }
```

Recorded source-to-declaration screening **Verdict:** `matches`
Reason: A source round combines the newly received applications with the old waiting list and retains the top q while rejecting the rest. The Lean function is that simultaneous update.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceStepRelation`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceStepRelation`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The proceduretерminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   THEOREM 2. Everyapplicant is at least as well offundertheassignmentgiven
by thedeferredacceptanceprocedureas he would be underany otherstableassign-
ment.

[Next verbatim source excerpt]

   DEFINITION. An assignment  ofapplicantstocolleges
                                                willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                  A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers

[Next verbatim source excerpt]

    4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
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when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   5. Optimality.We now show that the "deferredacceptance" procedurejust
described yields not only a stable but an optimal assignment of applicants.
That is,
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} [inst : Fintype Applicants] [inst_1 : Fintype Colleges]
  [inst_2 : DecidableEq Applicants] [inst_3 : DecidableEq Colleges] (quota : Colleges → ℕ)
  (val_applicant : Applicants → Colleges → ℝ) (val_college : Colleges → Applicants → ℝ)
  (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a')
  (before after : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges),
  after =
    GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceStep quota val_applicant val_college hcollegeStrict before
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: The source proceeds by repeated batches of applications, quota-sized waiting-list selection, and rejection. The Lean relation is exactly one such source procedure round.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.initialSourceState`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.initialSourceState`
Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim paper-source connection

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
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procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
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[Next verbatim source excerpt]

   THEOREM 2. Everyapplicant is at least as well offundertheassignmentgiven
by thedeferredacceptanceprocedureas he would be underany otherstableassign-
ment.

[Next verbatim source excerpt]

   DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                   A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers

[Next verbatim source excerpt]

    4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   5. Optimality.We now show that the "deferredacceptance" procedurejust
described yields not only a stable but an optimal assignment of applicants.
That is,
```

Lean-expanded paper semantic target

```
{Applicants : Type u_1} → {Colleges : Type u_2} → [inst : DecidableEq Applicants] → { applications := fun x => ∅ }
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: Before the first applications, no college has yet retained applicants. The Lean state is the empty-waiting-list initial state from which the stated procedure starts.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceReachable`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceReachable`
Source locator: source.txt:232-249

Verbatim paper-source connection

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} [inst : Fintype Applicants] [inst_1 : Fintype Colleges]
  [inst_2 : DecidableEq Applicants] [inst_3 : DecidableEq Colleges] (quota : Colleges → ℕ)
  (val_applicant : Applicants → Colleges → ℝ) (val_college : Colleges → Applicants → ℝ)
  (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a')
  (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges),
  Relation.ReflTransGen
    (GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceStepRelation quota val_applicant val_college
      hcollegeStrict)
    GS62CollegeAdmissions.ExactCollegeBatchedProcedure.initialSourceState s
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: The source defines the college procedure by successive application and waiting-list rounds from the empty initial state. The Lean predicate identifies exactly the states reachable by those rounds; it does not introduce an externally supplied terminal trace.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView`

Paper-local declaration:
`GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView`
Source locator: source.txt:232-249

Verbatim paper-source connection

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The proceduretерminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.
```

Lean-expanded paper semantic target

```
{Applicants : Type u_1} →
  {Colleges : Type u_2} →
    [inst : Fintype Applicants] →
      [inst_1 : Fintype Colleges] →
        [inst_2 : DecidableEq Applicants] →
          [inst_3 : DecidableEq Colleges] →
            (quota : Colleges → ℕ) →
              (val_college : Colleges → Applicants → ℝ) →
                (hcollegeStrict : ∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') →
                  (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges) →
                    {
                      app_match :=
                        GS62CollegeAdmissions.ExactCollegeBatchedProcedure.assignedCollege quota val_college
                          hcollegeStrict s,
                      college_roster := fun c =>
                        {a |
                          GS62CollegeAdmissions.ExactCollegeBatchedProcedure.assignedCollege quota val_college
                              hcollegeStrict s a =
                            some c},
                      consistent := ⋯ }
```

Recorded source-to-declaration screening Verdict: `matches`

Reason: The source assigns each applicant to the college whose final waiting list contains them, and leaves otherwise rejected applicants unassigned. The Lean view reads exactly that assignment from the terminal waiting-list state.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`GS62CollegeAdmissions.gs62ReplacementPair`

Paper-local declaration:
`GS62CollegeAdmissions.gs62ReplacementPair`
Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim paper-source connection

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                  willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                    A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  ∃ alpha beta A B,
    mu.app_match alpha = some A ∧
      mu.app_match beta = some B ∧
        val_applicant beta B < val_applicant beta A ∧ val_college A alpha < val_college A beta
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: The source definition of an unstable college assignment is the existence of the stated applicant-college replacement pair. The Lean predicate gives exactly those pair conditions.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.gs62LiteralStableCollegeAssignment

Paper-local declaration:
GS62CollegeAdmissions.gs62LiteralStableCollegeAssignment
Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim paper-source connection

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                   A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  ¬GS62CollegeAdmissions.gs62ReplacementPair val_applicant val_college mu
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source calls a college assignment stable exactly when it has no replacement-pair instability. The Lean predicate is the corresponding negation of that source definition.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

`GS62CollegeAdmissions.gs62LiteralApplicantOptimalCollegeAssignment`

Paper-local declaration:
`GS62CollegeAdmissions.gs62LiteralApplicantOptimalCollegeAssignment`
Source locator: source.txt:115-117 (printed page 10, second displayed Definition)

Verbatim paper-source connection

```
DEFINITION. A stableassignment                      applicantis at least
                              is calledoptimalif every
as welloffunderit as underanyotherstableassignment.
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.gs62LiteralStableCollegeAssignment val_applicant val_college mu ∧
    ∀ (nu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
      GS62CollegeAdmissions.gs62LiteralStableCollegeAssignment val_applicant val_college nu →
        ∀ (a : Applicants),
          EconCSLib.Matching.ManyToOne.valApplicant val_applicant a (nu.app_match a) ≤
            EconCSLib.Matching.ManyToOne.valApplicant val_applicant a (mu.app_match a)
```

Recorded source-to-declaration screening **Verdict:** `matches`

Reason: The source calls an assignment optimal for applicants when every applicant weakly prefers it to every other stable assignment. The Lean predicate is that literal comparison.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GS62CollegeAdmissions.PaperInterface.literalApplicantOptimalCollegeAssignment

Paper-local declaration:
GS62CollegeAdmissions.PaperInterface.literalApplicantOptimalCollegeAssignment
Source locator: source.txt:115-117 (printed page 10, second displayed Definition)

Verbatim paper-source connection

```
DEFINITION. A stableassignment                      applicantis at least
                               is calledoptimalif every
as welloffunderit as underanyotherstableassignment.
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.gs62LiteralApplicantOptimalCollegeAssignment val_applicant val_college mu
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The source's second displayed definition selects, among its literal stable assignments, one that every applicant weakly prefers to every other such assignment. The Lean declaration writes that source comparison class and applicant-by-applicant ordering directly.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.PaperInterface.replacementPairCollegeInstability

Paper-local declaration:
GS62CollegeAdmissions.PaperInterface.replacementPairCollegeInstability
Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim paper-source connection

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                   A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.gs62ReplacementPair val_applicant val_college mu
```

Recorded source-to-declaration screening Verdict: matches

Reason: The displayed source definition identifies instability through an applicant who prefers another college and a replacement at that college whom it ranks lower. The Lean predicate is that replacement pair.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.PaperInterface.unstableCollegeAssignment

Paper-local declaration:
GS62CollegeAdmissions.PaperInterface.unstableCollegeAssignment
Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim paper-source connection

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                   A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Lean-expanded paper semantic target

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.PaperInterface.replacementPairCollegeInstability val_applicant val_college mu
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source's first displayed college definition is the existence of two currently assigned applicants and colleges for which one applicant prefers the other college and that college prefers the replacement. The Lean predicate is exactly that replacement-pair condition.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

GS62CollegeAdmissions.PaperInterface.unstableMarriage

Paper-local declaration:
GS62CollegeAdmissions.PaperInterface.unstableMarriage
Source locator: source.txt:133-139 (printed page 11, Section 3)

Verbatim paper-source connection

```
A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Lean-expanded paper semantic target

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) (mu : EconCSLib.Matching.Assignment M W),
  ((∀ (m : M), ∃ w, mu.m_match m = some w) ∧ ∀ (w : W), ∃ m, mu.w_match w = some m) ∧
    ∃ m w,
      EconCSLib.Matching.valM val_m m (mu.m_match m) < val_m m w ∧
        EconCSLib.Matching.valW val_w w (mu.w_match w) < val_w w m
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source calls a complete marriage unstable when some man and woman each prefer one another to their current partners. The Lean predicate records the same complete matching and mutual strict improvement condition.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Main-text source claims

Review row 1

Source locator: source.txt:93-98 (printed page 10, first displayed Definition)

Verbatim source input

```
DEFINITION. An assignment  ofapplicantstocolleges
                                                 willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                   A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers
```

Expanded PaperInterface specification

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.PaperInterface.unstableCollegeAssignment val_applicant val_college mu ↔
    ∃ alpha beta A B,
      mu.app_match alpha = some A ∧
        mu.app_match beta = some B ∧
          val_applicant beta B < val_applicant beta A ∧ val_college A alpha < val_college A beta
```

Lean proof endpoint (built separately)

```
GS62CollegeAdmissions.ProofInterface.unstableCollegeAssignment_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `matches`

Reason: The transparent Spec states the displayed current-assignment replacement pair directly: two currently assigned applicants, their current colleges, the applicant's strict improvement, and the college's strict preference for the replacement applicant. It adds no vacancy, unmatched-applicant, or individual-rationality condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: source.txt:115-117 (printed page 10, second displayed Definition)

Verbatim source input

```
DEFINITION. A stableassignment                      applicantis at least
                            is calledoptimalif every
as welloffunderit as underanyotherstableassignment.
```

Expanded PaperInterface specification

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} (val_applicant : Applicants → Colleges → ℝ)
  (val_college : Colleges → Applicants → ℝ) (mu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  GS62CollegeAdmissions.PaperInterface.literalApplicantOptimalCollegeAssignment val_applicant val_college mu ↔
    (¬∃ alpha beta A B,
          mu.app_match alpha = some A ∧
            mu.app_match beta = some B ∧
              val_applicant beta B < val_applicant beta A ∧ val_college A alpha < val_college A beta) ∧
      ∀ (nu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
        (¬∃ alpha beta A B,
              nu.app_match alpha = some A ∧
                nu.app_match beta = some B ∧
                  val_applicant beta B < val_applicant beta A ∧ val_college A alpha < val_college A beta) →
          ∀ (a : Applicants),
            (match nu.app_match a with
              | none => 0
              | some c => val_applicant a c) ≤
              match mu.app_match a with
              | none => 0
              | some c => val_applicant a c
```

Lean proof endpoint (built separately)

```
GS62CollegeAdmissions.ProofInterface.literalApplicantOptimalCollegeAssignment_iff_source_definition
```

Recorded source-to-Spec screening Verdict: `matches`

Reason: The transparent Spec preserves the literal page-10 comparison class: quota-feasible assignments with no displayed replacement pair and applicant-by-applicant weak preference for the selected assignment. This row intentionally does not use the Section 4 operational stability convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: source.txt:133-139 (printed page 11, Section 3)

Verbatim source input

```
A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) (mu : EconCSLib.Matching.Assignment M W),
  GS62CollegeAdmissions.PaperInterface.unstableMarriage val_m val_w mu ↔
    ((∀ (m : M), ∃ w, mu.m_match m = some w) ∧ ∀ (w : W), ∃ m, mu.w_match w = some m) ∧
      ∃ m w,
        EconCSLib.Matching.valM val_m m (mu.m_match m) < val_m m w ∧
          EconCSLib.Matching.valW val_w w (mu.w_match w) < val_w w m
```

Lean proof endpoint (built separately)

```
GS62CollegeAdmissions.ProofInterface.unstableMarriage_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `matches`

Reason: The source's marriage discussion concerns a complete matching and a man-woman pair who each prefer the other to the present partner. The transparent Spec writes the completeness and both strict present-partner comparisons explicitly.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: source.txt:127-139; source.txt:185 (printed pages 11-12, Section 3 and Theorem 1)

Verbatim source input

```
3. Stable assignments and a marriage problem. In trying to settle the
question of the existence of stable assignmentswe were led to look firstat a
special case, in whichthereare the same numberof applicants as colleges and all
quotas are unity. This situation is, of course, highlyunnatural in the context
of college admissions, but there is another "story" into which it fits quite
readily.
   A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.

[Next verbatim source excerpt]

     THEOREM 1. Therealwaysexistsa stablesetofmarriages.

[Next verbatim source excerpt]

   3. Stable assignments and a marriage problem. In trying to settle the
question of the existence of stable assignmentswe were led to look firstat a
special case, in whichthereare the same numberof applicants as colleges and all
quotas are unity. This situation is, of course, highlyunnatural in the context
of college admissions, but there is another "story" into which it fits quite
readily.

[Next verbatim source excerpt]

   A certain communityconsists of n men and n women. Each person ranks
those of the opposite sex in accordance with his or her preferencesfora marriage
partner.We seek a satisfactoryway of marryingoffall membersof the com-
munity.Imitatingour earlierdefinition,we call a set of marriagesunstable(and
here the suitabilityof the termis quite clear) if under it thereare a man and a
woman who are not marriedto each other but prefereach other to theiractual
mates.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [DecidableEq M] [DecidableEq W]
  (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Fintype.card M = Fintype.card W →
    (∀ (m : M) (w w' : W), val_m m w = val_m m w' → w = w') →
      (∀ (w : W) (m m' : M), val_w w m = val_w w m' → m = m') →
        (∀ (m : M) (w : W), 0 < val_m m w) →
          (∀ (w : W) (m : M), 0 < val_w w m) →
            ∃ mu,
              ¬(((∀ (m : M), ∃ w, mu.m_match m = some w) ∧ ∀ (w : W), ∃ m, mu.w_match w = some m) ∧
                    ∃ m w,
                      EconCSLib.Matching.valM val_m m (mu.m_match m) < val_m m w ∧
                        EconCSLib.Matching.valW val_w w (mu.w_match w) < val_w w m) ∧
                (∀ (m : M), ∃ w, mu.m_match m = some w) ∧ ∀ (w : W), ∃ m, mu.w_match w = some m
```

Lean proof endpoint (built separately)

```
GS62CollegeAdmissions.ProofInterface.theorem1_stable_marriage_exists
```

Recorded source-to-Spec screening Verdict: `matches`

Reason: The transparent Spec has the source's finite equal-size strict complete domain and concludes a complete assignment with no mutually preferred blocking pair. Its expanded form makes the source's stable-marriage claim checkable without a wrapper predicate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: source.txt:232-249; source.txt:253-255 (printed pages 13-14, Sections 4-5 and Theorem 2)

Verbatim source input

```
4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   THEOREM 2. Everyapplicant is at least as well offundertheassignmentgiven
by thedeferredacceptanceprocedureas he would be underany otherstableassign-
ment.

[Next verbatim source excerpt]

   DEFINITION. An assignment  ofapplicantstocolleges
                                                willbecalledunstableif
thereare twoapplicantsa and ,Bwhoare assignedtocolleges
                                                    A andB, respectively,
although        A toB and A prefers: toa.
        ,3prefers

[Next verbatim source excerpt]

    4. Stable assignments and the admissions problem. The extension of our
"deferred-acceptance"procedureto the problemofcollegeadmissionsis straight-
forward.For conveniencewe will assume that ifa college is not willingto accept
a student underany circumstances,as describedin Section 2, then that student
will not even be permittedto apply to the college. With this understandingthe
procedure follows: First, all students apply to the college of their firstchoice.
A college with a quota of q then places on its waiting list the q applicants who
rank highest,or all applicants if there are fewerthan q, and rejects the rest.
Rejected applicants then apply to their second choice and again each college
selects the top q fromamong the new applicants and those on its waiting list,
puts these on its new waitinglist,and rejects the rest.The procedureterminates
when every applicant is either on a waiting list or has been rejected by every
college to which he is willingand permittedto apply. At this point each college

[form-feed in source extraction]
14             COLLEGE ADMISSIONS AND STABILITY OF MARRIAGE              [January

admits everyoneon its waitinglist and the stable assignmenthas been achieved.
The proofthat the assignmentis stable is entirelyanalogous to the proofgiven
forthe marriageproblemand is leftto the reader.

[Next verbatim source excerpt]

   5. Optimality.We now show that the "deferredacceptance" procedurejust
described yields not only a stable but an optimal assignment of applicants.
That is,
```

Expanded PaperInterface specification

```
∀ {Applicants : Type u_1} {Colleges : Type u_2} [inst : Fintype Applicants] [inst_1 : Fintype Colleges]
  [inst_2 : DecidableEq Applicants] [inst_3 : DecidableEq Colleges] (quota : Colleges → ℕ)
  (val_applicant : Applicants → Colleges → ℝ) (val_college : Colleges → Applicants → ℝ),
  ((∀ (a : Applicants) (c c' : Colleges), val_applicant a c = val_applicant a c' → c = c') ∧
     ∀ (a : Applicants) (c : Colleges), val_applicant a c ≠ 0) →
    ∀
      (hcollege_strict :
        (∀ (c : Colleges) (a a' : Applicants), val_college c a = val_college c a' → a = a') ∧
          ∀ (c : Colleges) (a : Applicants), val_college c a ≠ 0),
      (∃ s,
          GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceReachable quota val_applicant val_college ⋯ s ∧
            ¬∃ a,
                GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceActive quota val_applicant val_college ⋯ s a) ∧
        ∀ (s : GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceWaitingListState Applicants Colleges),
          GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceReachable quota val_applicant val_college ⋯ s →
            (¬∃ a,
                  GS62CollegeAdmissions.ExactCollegeBatchedProcedure.SourceActive quota val_applicant val_college ⋯ s
                    a) →
              ((∀ (c : Colleges),
                    ((GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota val_college ⋯
                              s).college_roster
```

```
          c).card ≤
        quota c) ∧
    (∀ (a : Applicants),
        0 ≤
          EconCSLib.Matching.ManyToOne.valApplicant val_applicant a
            ((GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota val_college ⋯
                s).app_match
              a)) ∧
      (∀ (c : Colleges),
          ∀
            a ∈
              (GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota val_college ⋯
                    s).college_roster
                c,
              0 ≤ val_college c a) ∧
          ∀ (a : Applicants) (c : Colleges),
            EconCSLib.Matching.ManyToOne.valApplicant val_applicant a
                  ((GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota
                      val_college ⋯ s).app_match
                    a) <
                val_applicant a c →
              (0 < val_college c a ∧
                    ((GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota
                              val_college ⋯ s).college_roster
                          c).card <
                      quota c ∨
                  ∃
                    a' ∈
                      (GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota
                            val_college ⋯ s).college_roster
                        c,
                    val_college c a' < val_college c a) →
                False) ∧
∀ (nu : EconCSLib.Matching.ManyToOneAssignment Applicants Colleges),
  ((∀ (c : Colleges), (nu.college_roster c).card ≤ quota c) ∧
      (∀ (a : Applicants),
          0 ≤ EconCSLib.Matching.ManyToOne.valApplicant val_applicant a (nu.app_match a)) ∧
        (∀ (c : Colleges), ∀ a ∈ nu.college_roster c, 0 ≤ val_college c a) ∧
          ∀ (a : Applicants) (c : Colleges),
            EconCSLib.Matching.ManyToOne.valApplicant val_applicant a (nu.app_match a) <
                val_applicant a c →
              (0 < val_college c a ∧ (nu.college_roster c).card < quota c ∨
                  ∃ a' ∈ nu.college_roster c, val_college c a' < val_college c a) →
                False) →
    ∀ (a : Applicants),
      EconCSLib.Matching.ManyToOne.valApplicant val_applicant a (nu.app_match a) ≤
        EconCSLib.Matching.ManyToOne.valApplicant val_applicant a
          ((GS62CollegeAdmissions.ExactCollegeBatchedProcedure.sourceAssignmentView quota val_college ⋯
                s).app_match
            a)
```

Lean proof endpoint (built separately)

`GS62CollegeAdmissions.ProofInterface.theorem2_applicant_optimality`

Recorded source-to-Spec screening **Verdict:** `matches`

Reason: The raw bundle includes both the earlier displayed replacement-pair condition and Sections 4–5. Read together, the latter uses stable assignment for the terminal top-quota waiting-list outcome and says its stability proof is analogous to the marriage proof; it therefore supplies the completed operational comparison class used by the transparent Spec. The literal display remains a distinct source claim and is not silently identified with that Section 4–5 reading.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation